Review plan · for the lecturer

Teaching statistical learning at university: unit check

Data science · University · Unit check

Time	Materials
40 min	Projector or board, and the lecture slides.

Learning goals

- Sampling and the bootstrap
- Bias, variance and CV
- Causal inference
- Module design

Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

Discussion

- Which misconception will you target first?
- Which lab will you run this term?
- Which assessment will you change?

Exit ticket

- One change I will make to my module this term, and how I will know it worked.

Try it in your next lecture
Bring your redesigned alignment table to the next department meeting and ask one colleague to review it.

Quiz sheet · for students

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Name _____ Group _____ Date _____

1 **σ = 20 and n = 100. What is the standard error of the mean?**

◯ 10 ◯ 0.2 ◯ 20 ◯ 2

2 **A 95% CI of 60–68 means P(60 < μ < 68) = 0.95.**

◯ True ◯ False

3 **Which are limits of the bootstrap? Choose all that apply.**

☐ It cannot fix a biased sample ☐ It only works for the mean ☐ It is unreliable for the maximum

4 **Work out the CV error.**

Fold errors 6, 8, 7, 9, 10 give a CV error of _____.

Word bank: 10 · 8 · 40

5 **Put a leak-free workflow in order.**

☐ Choose the model by CV error

☐ Hold out a final test set

☐ Report error once on the test set

☐ Inside each CV fold, fit preprocessing and model

6 **Match each variable to its role in the tutoring DAG.**

Prior grade	• • • •	Mediator
Hours studied	• • • •	Collider
Dean's list	• • • •	Confounder

7 **Treated: 30 → 38. Comparison: 32 → 35. The DiD estimate is:**

◯ 11 ◯ 5 ◯ 3 ◯ 8

8 **Adding more data mainly lowers a model's bias.**

◯ True ◯ False

9 **Which make a module better aligned? Choose all that apply.**

☐ An assessment for each outcome ☐ Outcomes with active, observable verbs

☐ Sharing the rubric only after grading

10 **A student's CV accuracy is 95%, but on the final test set it is 70%. The most likely cause is:**

◯ Too much data ◯ Leakage into the cross-validation, or tuning many choices on the same folds

◯ Too few features

Answers · for the lecturer

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1 **2**

20/√100 = 20/10 = 2.

2 **False**

The 95% describes the procedure over repeated samples, not this fixed interval.

3 **It cannot fix a biased sample; It is unreliable for the maximum**

It works for many statistics, but it can't create information the sample lacks.

4 **Fold errors 6, 8, 7, 9, 10 give a CV error of 8.**

40 / 5 = 8.

5 **Top to bottom: 3, 1, 4, 2**

The test set is touched once, at the end; everything else happens inside folds.

6 **Prior grade — Confounder**
Hours studied — Mediator
Dean's list — Collider

Adjust only for the confounder.

7 **5**

8 − 3 = 5.

8 **False**

More data mainly lowers variance; a too-simple model stays biased.

9 **Outcomes with active, observable verbs; An assessment for each outcome**

Rubrics teach only when students can use them while working.

10 **Leakage into the cross-validation, or tuning many choices on the same folds**

A large optimistic gap points to information leaking into model choice.